

Homework #5.

1a. Gain is a function of wavelength:

- Optical amplifiers, do not amplify all wavelengths equally. Their gain varies across the wavelength spectrum.

- Impact on multi-wavelength spectrum:

1. Some channels may be amplified more than others, leading to unequal received power at the destination.

2. Channel with lower gain may fall below the receiver sensitivity threshold, resulting bit errors.

3. To compensate, gain flattening filters or careful channel power management are required to maintain uniform signal quality across all wavelengths.

4. Wavelength assignment strategies might be influenced; wavelengths with lower amplifier gain may be spaced differently or transmitted with slightly higher input power.

b. Saturation of the amplifier gain:

- Optical amplifiers have a saturation power; does not proportionally increase output

In multiwave systems:

1. Gain saturation occurs when total input power is high, causing stronger channels to steal "gain" from weaker signals. This is called gain competition.

2. This can lead to uneven amplification, where some wavelengths are suppressed while others dominate.

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3. Network design considers power budgeting, ensuring the sum of all channel powers remain below the saturation limit.

4. It may require per-channel power control or distribute amplifiers to avoid signal degradation and maintain uniform signal across all wavelengths.

2) Given inputs
Transmitter

$$P_t = 1 \text{ mW}$$

$$\text{Extinction ratio: } \gamma = 100$$

$$x = 0$$

$$\text{Bit rate } B = 1 \text{ Gb/s}$$

$$\text{Pulse width: } T_0 = 0.05 \text{ ns}$$

$$\text{Wavelength spacing } \Delta\lambda = 0.1 \text{ nm}$$

Fiber:

$$\text{Attenuation: } \alpha = 0.25 \text{ dB/km}$$

$$\text{Dispersion } D = 17 \text{ ps/nm}$$

$$\text{Coupling/Splice loss: } 10 \text{ dB}$$

$$\text{Fiber length } = 50 \text{ km}$$

Receiver:

$$\text{Noise bandwidth } \Delta f = 2 \text{ GHz}$$

$$\text{Noise figure } F_n = 5$$

$$\text{Load resistance } R_L = 100 \Omega$$

$$\text{Temperature } T = 300 \text{ K}$$

$$\text{Responsivity } R = 10 \text{ A/W}$$

$$\text{Receiving power penalty: } PP_{\text{ran}} = 4 \text{ dB}$$

$$\text{Assumptions } \sigma_r^2 \geq \sigma_s^2 \text{ and } P_o = P_i/100$$

Step 1 Minimum receiver power for $BER = 10^{-9}$

Ideal Q factor for $BER = 10^{-9}$

$$BER = \frac{1}{2} \operatorname{erfc} \left(\frac{Q}{\sqrt{2}} \right)$$

For $BER = 10^{-9}$, $Q \approx 6$

Receiver thermal noise limited power

$$\sigma_T^2 = 4 \cdot k T R_L \Delta f \quad k = 1.38 \times 10^{-23} \text{ J/K}$$

$$T = 300 \text{ K}$$

$$R_L = 100 \Omega$$

$$\Delta f = 2 \text{ GHz} = 2 \times 10^9 \text{ Hz}$$

$$\sigma_T^2 = 4 \cdot 1.38 \times 10^{-23} \cdot 300 \cdot 100 \cdot 2 \times 10^9$$

$$= 5.52 \times 10^{-23} \cdot 300 \cdot 100 \cdot 2 \times 10^9$$

$$= 5.52 \times 1.636 \times 10^{-20} \cdot 100 \cdot 2 \times 10^9$$

$$= 1.656 \times 10^{-18} \cdot 2 \times 10^9$$

$$= 3.312 \times 10^{-9} \text{ A}$$

$$\sigma_T = \sqrt{3.312 \times 10^{-9} \text{ A}} = 1.820 \sqrt{10^{-9}}$$

$$\approx 5.756$$

$$= 1.820 \times 10^{-4.5}$$

$$\approx 5.76 \times 10^{-5} \text{ A}$$

Receiver current vs optical power.

$$I_{\text{signal}} = R \cdot P_{\text{rx}}$$

$$r = 100 \Rightarrow P_o / P_i = 1/100 = 0.01$$

$$Q = \frac{I_1 - I_0}{\sigma_T} = \frac{R(P_1 - P_0)}{\sigma_T} = \frac{RP_1(1 - 0.01)}{\sigma_T} \approx \frac{0.99RP_1}{\sigma_T}$$

$$6 = \frac{0.66 \cdot 10 \cdot P_{rx}}{5.756 \times 10^{-5}}$$

$$P_{rx} = \frac{6 \cdot 5.756 \times 10^{-5}}{0.66} \approx \frac{3.4536 \times 10^{-4}}{0.66}$$

$$P_{rx} \approx 3.49 \times 10^{-5} \text{ W.} \\ \approx 34.9 \mu\text{W}$$

b) Power penalties due to extinction ratio & dispersion.

1. Extinction ratio penalty (dB)

$$PP_r = 10 \log_{10} \left(\frac{r}{r-1} \right) = 10 \log_{10} \frac{100}{99} \approx 0.044 \text{ dB}$$

2. Dispersion penalty:

$$\Delta t = D_\lambda \cdot \Delta \lambda \cdot L$$

$$D_\lambda = 17 \text{ ps/nm}$$

$$\Delta \lambda = 0.1 \text{ nm}$$

$$L = 50 \text{ km}$$

$$\Delta t = 17 \cdot 0.1 \cdot 50 = 85 \text{ ps} = 0.085 \text{ ns.}$$

$$\text{Bit duration } T_b = 1/\mu = 1/1 \text{ Gb/s} = 1 \text{ ns.}$$

Dispersion penalty (dB)

$$PP_D \approx 10 \log_{10} \left(1 + \left(\frac{\Delta f}{f_b} \right)^2 \right) = 10 \log_{10} (1 + 0.085^2) \\ = 10 \log_{10} (1.0072) \\ \approx 0.03 \text{ dB}$$

c) Check if link can achieve $BER = 10^{-7}$ with 3 dB margin

• Total penalties $PP_{\text{total}} = PP_r + PP_D + 3$
 $\approx 0.044 + 0.03 + 3 \approx 3.074 \text{ dB}$

$$P_{rx(rq)} = P_{rx(min)} \cdot 10^{PP_{\text{total}}/10} \approx 35 \mu\text{W} \cdot 10^{3.074/10}$$

$$10^{3.074/10} = 10^{0.3074} \approx 2.03$$

$$P_{rx(\text{required})} \approx 35 \cdot 2.03 \approx 71 \mu\text{W}$$

Link power at receiver

$$P_{rx(\text{actual})} = P_t \cdot 10^{-(\alpha L + \text{coupling loss})/10} \\ = 1 \text{ mW} \cdot 10^{-(0.25 \times 20/10)} = 1 \cdot 10^{-(12.5 + 10)/10} \\ = 1 \cdot 10^{-2.25} \\ 10^{-2.25} \approx 0.00562 \text{ mW} = 5.62 \mu\text{W}$$

d) Using power budget method.

$$P_{rx} = P_{tx} - \text{fiber loss} - \text{coupling loss} - PP$$

$$\text{Total loss} = 12.5 + 10 + 3.074 = 25.574 \text{ dB}$$

$$P_{rx} = 0 \text{ dBm} - 25.574 \approx -25.574 \text{ dB}$$

P_r power: Convert to mW $P_r = 10^{-25.57/10}$
 $\approx 0.0028 \text{ mW} = 2.8 \mu\text{W}$

e) Importance of error margin

1. Aging of components
2. Environmental variations
3. Unexpected losses
4. Safety margin for system reliability

3) Given input

Modulation: NRZ

Bit duration $T_b = 1 \text{ ns} \rightarrow$ Bit rate $B = 1/T_b = 1 \text{ Gb/s}$

Chromatic dispersion $D = 10 \text{ ps/nm}$

Spectral width $\Delta\lambda = 1 \text{ nm}$

Fiber length $L = 10 \text{ km}$

Wavelength $\lambda_0 = 1.55 \mu\text{m}$

Transmitter rise time $T_{r,tx} = 0.1 \text{ ns}$

Receiver rise time $T_{r,rx} = 0.2 \text{ ns}$

Modal indices: $n_1 = 1.500$, $n_2 = 1.496$

$x = 0$

For NRZ, the total rise time

$$T_{r,sys} = \sqrt{T_{r,tx}^2 + T_{r,rx}^2 + T_{r,disp}^2 + T_{r,mod}^2 + T_{r,mod}^2}$$

Assume no modal dispersion yet.

$$T_{r,tx} = 0.1 \text{ ns}, T_{r,rx} = 0.2 \text{ ns}$$

$$T_{r,disp} = D \cdot \Delta\lambda \cdot L = 10 \text{ ps/nm} \cdot 1 \text{ nm} \cdot 10 \text{ km} = 100 \text{ ps} = 0.1 \text{ ns}$$

$$T_{r, \text{modal}} = \frac{n_1 - n_2}{c} \cdot L \cdot 10^3 \text{ km} \rightarrow \text{m}$$

$$L = 10,000 \text{ m.}$$

$$= \frac{0.004 \cdot 10,000}{3 \times 10^8} = \frac{40}{3 \times 10^8} = 1.33 \times 10^{-7} \text{ s}$$

$$= 133 \text{ ns}$$

Total system rise time

$$T_{r, \text{sys}} \approx \sqrt{0.1^2 + 0.2^2 + 0.1^2 + 133^2}$$

$$\approx \sqrt{17689.06} \approx 133 \text{ ns}$$

$$B_{\text{max}} \approx \frac{0.7}{T_{r, \text{sys}}} \approx \frac{0.7}{133 \text{ ns}} \approx 5.26 \text{ Mb/s}$$

$$B_{\text{max}} \approx 5.26 \text{ Mb/s.}$$

4) 1) WDM channels

a) Intra channel crosstalk.

It is possible because same wavelength light can leak between paths (imperfect). That leaked light is coherent with the desired signal & can therefore add destructively or constructively, produce a much larger effect than an equal power of a different wavelength.

2) Power Penalty

For $\lambda_3 \Rightarrow$ adjacent channels λ_2 & $\lambda_4 \Rightarrow 2 \times 0.01 = 0.02$.

Other $\Rightarrow \lambda_1 \Rightarrow 0.002 \times 1 = 0.002$.

$$\Sigma \epsilon_i = 0.02 + 0.002 = 0.022$$

Power Penalty.

$$\sqrt{\epsilon} = 0.1, \quad \frac{1 + \sqrt{\epsilon}}{1 - \sqrt{\epsilon}} = \frac{1.1}{0.9} = 1.22$$

$$PP_{\text{intra}} \approx 20 \log_{10} \left(\frac{1 + \sqrt{\epsilon}}{1 - \sqrt{\epsilon}} \right) = 20 \log_{10} (1.222) \approx 1.74 \text{ dB}$$

$$PP_{\text{total}} = PP_{\text{intra}} + PP_{\text{inter}} \approx 1.74 + 0.095 \approx 1.84 \text{ dB}$$

c) Would penalties be different for λ_1 edge vs λ_3 .

Yes, slightly - interchannel penalty changes, intrachannel penalty does not.

for λ_1 (edge channel)

Adjacent: only $\lambda_2 \rightarrow 1 \times 0.01 = 0.01$

Non-adjacent λ_3 & $\lambda_4 \rightarrow 2 \times 0.002 = 0.004$

Sum = 0.014

$$PP_{\text{inter}}(\lambda_1) = 10 \log_{10}(1.014) \approx 0.061 \text{ dB.}$$

interchannel penalty $\epsilon_{\text{inter}} = 0.01$ remains $\approx 1.74 \text{ dB.}$

$$\lambda \approx 1.74 + 0.061 \approx 1.80 \text{ dB.}$$

Thus, edge channel $\approx 1.80 \text{ dB}$, middle channel $\lambda_3 \approx 1.84 \text{ dB.}$

few tenths of a dB difference because interchannel contributions are small compared to the coherent interchannel term.